Secure Agentic Development

Evolving from Manual Coder to Mission Control Architect with Google Antigravity.

GOOGLE ANTIGRAVITY

INTRODUCTION

What is Antigravity?

An agent-first development platform that transforms the IDE into a goal-directed system capable of planning, coding, and autonomous verification.

Agentic Architecture Focus

Autonomous Planning

Agents don't just complete lines; they ingest entire codebases (2M token window) to map dependencies and formulate multi-step implementation strategies before writing a single line of code.

Multi-Surface Operation

One environment, three surfaces. Agents operate simultaneously across the Editor (code), the Terminal (execution), and the Browser (verification) to provide end-to-end task completion.

The Three Core Surfaces

The Editor

A familiar IDE for synchronous pair-programming, inline commands, and micro-edits with AI assistance.

Agent Manager

The Mission Control dashboard to spawn, monitor, and interact with multiple asynchronous agents.

The Browser

Integrated Chrome agent for autonomous UI testing, web scraping, and visual verification recordings.

Intelligence & Context

Gemini 3 Pro Power

Leveraging a massive context window to process real-world engineering challenges. Antigravity understands your project's unique rules and architectural patterns.

- 2,000,000 Token Context Window
- 76.2% SWE-bench Verified Performance
- 1487 Elo WebDev Arena Score

Unmatched Developer Velocity

10X

VELOCITY INCREASE

Beyond Auto-Complete

Traditional coding assistants reduce typing time. Antigravity reduces **cognitive load** by automating research, debugging, and verification—tasks that usually consume 80% of a developer's day.
Agents handle parallel processing for development tasks, allowing you to focus on high-level architecture while AI handles the execution loop.

"Antigravity closes the trust gap by providing tangible proof of work through Artifacts, allowing you to monitor high-level summaries before ever granting execution privileges to the machine."

— GOOGLE ANTIGRAVITY ENGINEERING TEAM

Flexible Installation Tiers

Tier	Platform	Strategy	Security Level
Tier 1: Easy	Mac / Win / Linux	Direct Binary Install	Standard
Tier 2: Medium	Mac / Linux	Homebrew Management	Controlled
Tier 3: Advanced	Docker / Podman	Containerized Sandbox	High (Isolated)
Lab: bluefin-wtg	Universal Blue	Zero-Drift AI Factory	Maximum (Immutable)

Trust Through Artifacts

Implementation Plans

Pre-flight architectural review before code generation.

Code Diffs

Line-by-line transparency for every change proposed.

Visual Walkthroughs

Screen recordings of agents verifying UI behavior.

The Agentic Workflow

PLAN

Agent generates architecture & Markdown task list.

EXECUTE

Review precise Code Diffs and tool calls.

VERIFY

Autonomous browser tests & recordings.

DEPLOY

Artifact review, human approval, and commit.

Built-In Security Guardrails

Terminal Permissions

Strictly define allowable actions. Force agents to request human review before running potentially dangerous bash scripts on local systems.

Secure Mode

Enable isolated execution to restrict agent access to sensitive external network resources and ensure perimeter safety.

Workspace Sandboxing

Deploy using containerized workflows (Tier 3) to prevent agents from altering the host OS during experimentation.

Ready to Begin?

Next Step: Module 2 Installation & Setup.

Let's Start Lab 1

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Course Overview

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What is Google Antigravity?

- **Mission Control:** An orchestration layer to spawn, monitor, and interact with multiple AI agents working asynchronously.
- **Agent-First Platform:** Evolves the traditional IDE into a goal-directed system capable of planning, executing, and autonomously verifying complex tasks.
- **Continuous Autonomy:** Watch agents write code, execute bash commands, and run browser checks natively inside one ecosystem.

Three Core Surfaces

The Editor

A familiar, high-performance IDE environment for synchronous pair-programming and micro-editing with the AI.

Agent Manager

The orchestration dashboard where you prompt, review, and strictly manage multiple asynchronous digital employees.

The Browser

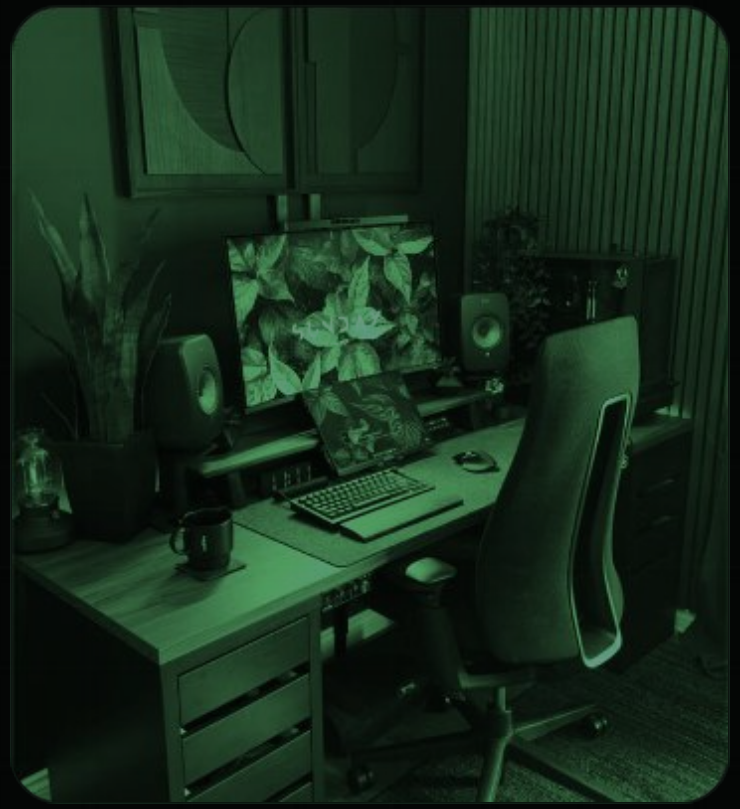
Native browser integration allowing agents to autonomously test User Interfaces and dynamically scrape target data.

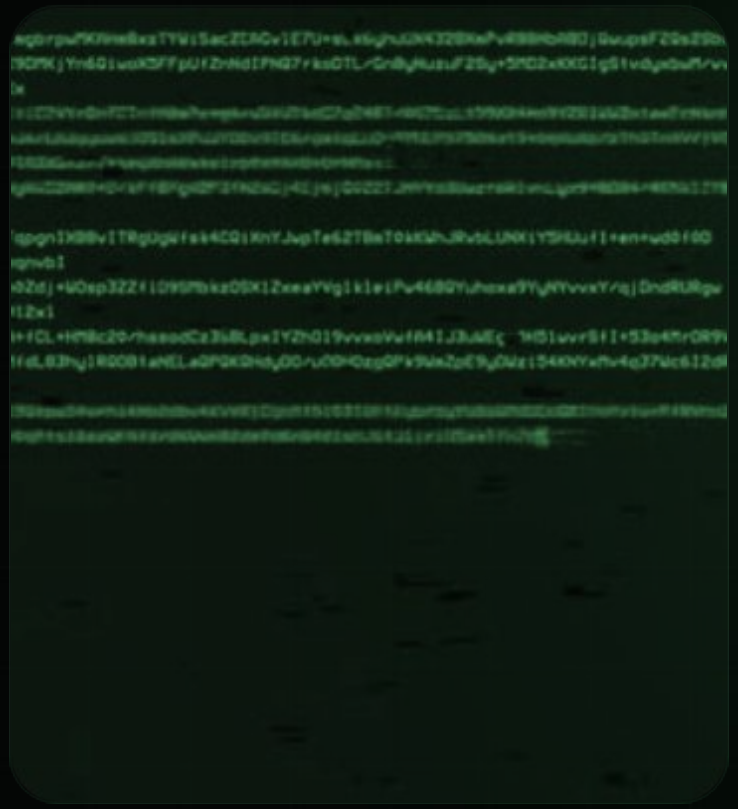
Bridging the Trust Gap

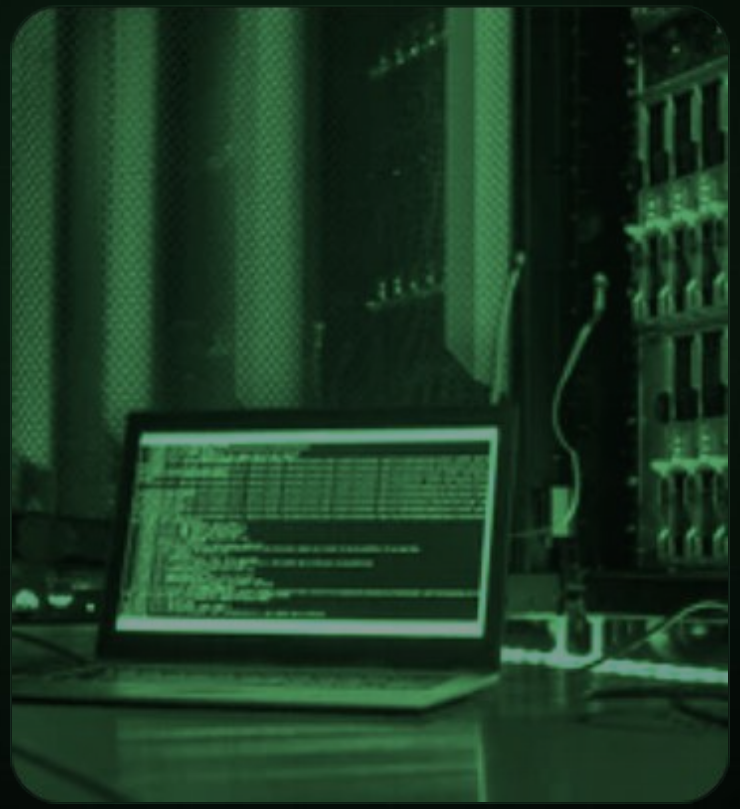
Traditional AI tools offer "Black Box" code generation. When dealing with autonomous agents capable of terminal execution, blindness is a massive security risk.

Antigravity closes the trust gap by providing **tangible proof of work** through Artifacts, allowing you to monitor high-level summaries before ever granting execution privileges to the machine.

Flexible Installation







Tier 1: Easy

Fastest setup for personal dev machines using a simple, direct binary install script.

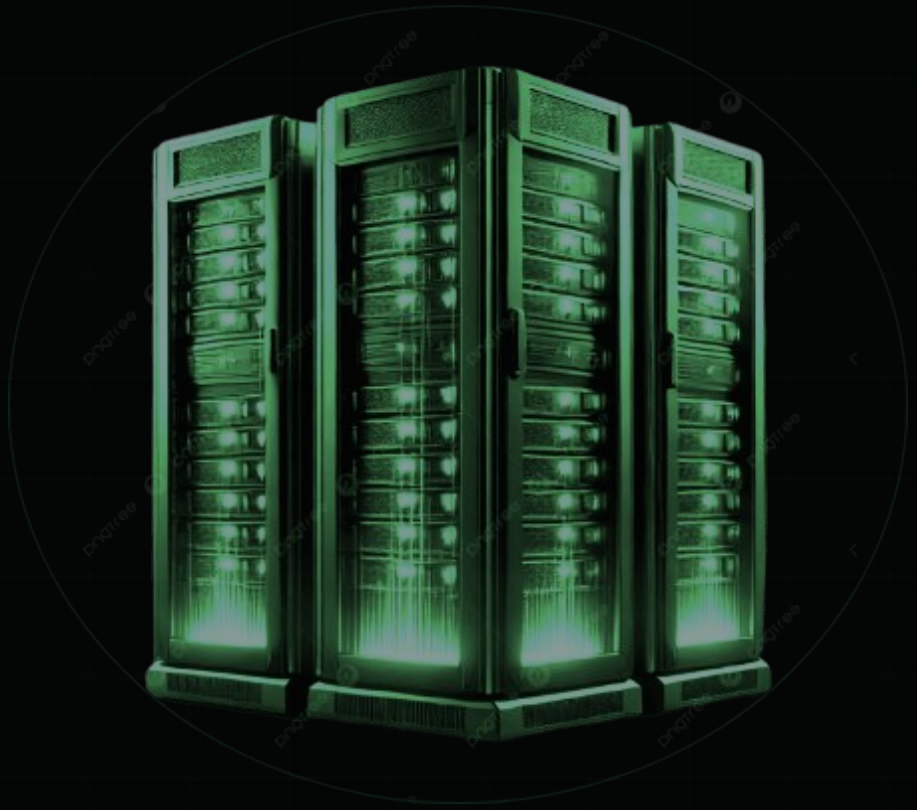
Tier 2: Brew

Standardized dependency management via Homebrew for MacOS/Linux professionals.

Tier 3: Advanced

Recommended. Full containerized isolation via Docker or Podman for high-risk labs.

The bluefin-wtg Lab



Zero-Drift AI Factory

The ultimate safety net. A fully immutable, isolated laboratory pre-loaded with everything you need to run Vision-Language Models (VLMs).

It eliminates dependency hell by enforcing a strict one-repository-to-one-artifact rule, securing the host OS from dangerous terminal executions.

Security Guardrails

>_ Terminal Permissions

Antigravity allows you to strictly define what an autonomous agent is allowed to do. Force the agent to request human review before running any potentially dangerous bash scripts on your local system.

Secure Mode

Enhance your overall perimeter security by enabling "Secure Mode", which strictly restricts the agent's access and prevents outbound connections to sensitive external network resources.

The Agentic Workflow

EXECUTE

Review precise Code Diffs and monitor terminal tool calls in real-time.

DEPLOY

Final artifact review, human approval sign-off, and codebase commit.

PLAN

Agent generates architecture and produces a structured

VERIFY

Observe autonomous browser UI tests and generate

GOOGLE ANTIGRAVITY

Providing Feedback

Planning Phase: Review the initial architecture implementation plan and leave comments to change the tech stack entirely before code is written.

Task List Phase: Add strict comments specifying the exact testing criteria you want the agent to follow to reach validation.

Coding Phase: Quickly accept, reject, or comment on inline code refactors seamlessly within the Editor's diff view.

Verification Phase: Request visual or UI changes by commenting directly on the browser recordings produced by the agent.

Rules vs. Workflows

Persistent Rules

Global guidelines the agent must always follow automatically in the background. Codify corporate standards by enforcing PEP 8 styling, or mandating that strict corporate copyright headers be applied to every file.

On-Demand Workflows

Saved, macro-style prompts that are only triggered via slash-commands when you specifically need them. Quickly execute repetitive tasks like typing /generate-unit-tests for your active file.

Disaster Recovery

100%

Ephemeral Safety

The "Panic Button"

If you deploy using the Advanced Tier 3 Sandbox, you possess the ultimate safety switch. Should an agent hallucinate and delete critical system binaries, simply exit the sandbox. The corrupted state vanishes, and a perfect, pristine environment is instantly rebuilt from the local cache.

Ready to Begin?

Next Step: Module 2 Installation & Setup.

Let's Start Lab 1